Automotive Software Should-Cost

Premium Luxury SUV — Full Software Stack (2024–2026)

Engineering hub: United Kingdom · Source: Tier1_Supplier · Life: 6yr · Volume: 200k/yr

Generated: 14/10/2026 · CostVision

£304.6M

Total Programme Cost

£254 / vehicle

1. Programme Cost Summary

Cost Bucket	Value (£M)	Share (%)
Development Engineering	£28.7M	9.4%
Testing & Validation	£11.8M	3.9%
Integration & V&V	£5.3M	1.7%
Toolchain (dev tools)	£15.9M	5.2%
Calibration & Tuning	£2.4M	0.8%
Cybersecurity	£1.5M	0.5%
Maintenance (lifecycle)	£34.0M	11.2%
Cloud & Infra	£103.9M	34.1%
IP Licensing	£97.6M	32.0%
NRE Subtotal	£69.2M	22.7%
TOTAL	£304.6M	100%

Rate basis

Rate book	CostVision built-in v2.0.0 (country book 2026-10) (reviewed 2026-10)
Engineering hub	United Kingdom — its own salaries, statutory employer on-costs and working hours (no UK multiplier)
Hub rate (£ / person-month of 152 h, before overhead)	mid £6,237 · senior £6,531 — GBP 66,875 mid / 70,000 senior gross a year + employer on-cost 13.9% at mid = 76,156 / 79,750; ÷ 1,856 productive h × 152 h = GBP 6,237 / 6,531 per person-month ÷ 1 = £6,237 / £6,531
Engineer rate used	£6,399 — 55 % senior blend of the hub's mid and senior rates
Development source	Tier1_Supplier × 0.9966024915062288
Engineering burden	× 1.610 = employment +15 % × company +40 % × capital +0 % (on the hub rate)
Loaded rate (£ / person-month)	£10,267
Rate confidence	Medium — figures read from search results of the cited pages (Oct 2026); see the rate book
Powertrain	PHEV
Platform annual volume	not shared (this programme carries all of it)
Annual volume × life	200,000 × 6 yr
NRE recovery	6 yr
Discount rate	0 %

Development duration (tool licences)	54 months
Cloud	per connected vehicle-year (module £/yr ÷ 440,000 reference fleet) × 100 % connected, each vehicle served 8 yr; fixed platform £0 / yr
Map / per-vehicle-year royalties	3 yr per vehicle
Development tools	£39,500 per engineer-seat-year
Maintenance profile	full 3 yr after SOP, then 50 %; wage escalation 0 % / yr
Security support after the last build	10 yr at 25 % of the tail team — After the last build: each module's tail sustaining team × 25 % for 10 years — the duty is regulatory (UN R155: the post-production phase runs "until the end-of-life of all vehicles under the vehicle type"; R156: updates over the vehicles' lifetime); 10 years = Volkswagen / CARIAD's stated software support after end of production (2023, via the press); 25 % is a CostVision assumption — no published cost.
Homologation	on — organisation CSMS / SUMS shared by 1 vehicle type(s)
Nominal effort basis	53 catalogue · 0 COCOMO II size (2.94 × KSLOC^1.0997) · 0 custom PM
Effort calibration	none (model as published)

2. Monte Carlo Cost Distribution

Percentile	Total Cost (£M)	£/Vehicle
P10 (Optimistic)	£293.1M	£244
P50 (Median)	£346.0M	£288
P90 (Pessimistic)	£410.0M	£342
Mean	£349.3M	
Headline (~ P16)	£304.6M	£254

3. Cost by Software Category

Category	Modules	Grand Total (£M)	Share (%)
Powertrain & Battery	13	£31.0M	10.2%
ADAS Level 2 & 2+	9	£71.2M	23.4%
Infotainment, Connectivity & UX	9	£116.9M	38.4%
Vehicle Domain Controllers	7	£14.5M	4.8%
Middleware & Platform	6	£8.1M	2.7%
Cybersecurity (ISO 21434)	5	£11.3M	3.7%
OTA & Cloud Backend	4	£47.7M	15.7%

4. Module Cost Detail (Top 20 by Cost)

Module	Cat	ASIL	PM	Dev (£M)	Calib (£M)	Test (£M)	Total (£M)	£/Veh
Navigation Stack	C	QM	30	£0.3M	£0.0M	£0.1M	£50.6M	£42
IVI Operating System (An…	C	QM	144	£1.5M	£0.0M	£0.5M	£31.1M	£26
Cloud Backend Services	G	QM	58	£0.6M	£0.0M	£0.2M	£29.3M	£24
Camera Perception Stack	B	B	163	£1.7M	£0.1M	£0.8M	£26.2M	£22
Telematics Control Unit …	C	B	30	£0.3M	£0.0M	£0.1M	£21.2M	£18
Sensor Fusion Engine	B	C	136	£1.4M	£0.1M	£0.7M	£10.2M	£9

Module	Cat	ASIL	PM	Dev (£M)	Calib (£M)	Test (£M)	Total (£M)	£/Veh
Highway Assist / Traffic…	B	C	108	£1.1M	£0.1M	£0.6M	£9.6M	£8
Path Planning Algorithms	B	C	115	£1.2M	£0.1M	£0.6M	£8.2M	£7
Data Pipeline & Telemetry	G	QM	22	£0.2M	£0.0M	£0.1M	£8.1M	£7
Driver Monitoring Softwa…	B	B	69	£0.7M	£0.0M	£0.3M	£7.0M	£6
Vehicle Voice Assistant	C	QM	58	£0.6M	£0.0M	£0.2M	£6.8M	£6
OTA Update Manager (Vehi…	G	B	30	£0.3M	£0.0M	£0.1M	£5.3M	£4
Fleet Management & Analy…	G	QM	12	£0.1M	£0.0M	£0.0M	£5.0M	£4
Radar Processing Stack	B	B	75	£0.8M	£0.1M	£0.3M	£4.6M	£4
SOC/SOH/SOE Estimation M…	A	C	48	£0.5M	£0.1M	£0.2M	£4.5M	£4
Vehicle Motion Managemen…	D	C	115	£1.2M	£0.2M	£0.5M	£4.4M	£4
Hybrid Supervisory Contr…	A	C	115	£1.2M	£0.2M	£0.5M	£4.4M	£4
BMS Core Software	A	D	117	£1.2M	£0.1M	£0.5M	£4.3M	£4
Secure OTA Software Upda…	F	B	20	£0.2M	£0.0M	£0.1M	£3.6M	£3
AUTOSAR Adaptive Platform	E	B	60	£0.6M	£0.0M	£0.2M	£3.4M	£3

5. Sensitivity Analysis

Parameter	Low	Base	High	Range
ASIL level (all modules: D vs B)	£304.1M	£304.6M	£334.3M	£30.2M
Complexity (all modules: Medium vs Very High)	£279.0M	£304.6M	£315.6M	£36.7M
Reuse level (Heavy vs Fresh)	£265.0M	£304.6M	£370.9M	£105.9M
Engineering hub (Egypt vs USA — California)	£228.2M	£304.6M	£378.0M	£149.8M
Programme life (8 vs 14 years)	£377.6M	£304.6M	£596.8M	£219.1M
Effort catalogue (× 0.5 vs × 2)	£254.8M	£304.6M	£404.1M	£149.3M
Company burden (20 % vs 60 %)	£292.6M	£304.6M	£316.5M	£23.9M
Connected-service life (5 vs 15 years)	£265.6M	£304.6M	£395.4M	£129.8M
Security support after the last build (0 vs 15 years)	£297.2M	£304.6M	£308.3M	£11.1M
Map subscription (1 vs 7 years)	£277.0M	£304.6M	£359.8M	£82.8M
Production Volume (150k vs 50k units/yr, per-vehicle)	£285	£254	£539	£253

6. Evidence behind the total

Evidence grade	In the total	Share	Band drawn (low · high)
Your figures	£0.0M	0%	x 0.9 · x 1.15
Calibrated to your actuals	£0.0M	0%	x 0.85 · x 1.25
Published source	£22.7M	7%	x 0.85 · x 1.2
Labelled estimate	£46.2M	15%	x 0.7 · x 1.6
Unsourced catalogue	£235.7M	77%	x 0.6 · x 1.9

Reuse: 53 of 53 modules at the default Medium (x 0.60); written fresh, engineering would cost £43.7M more. Largest unsourced: Cloud (per connected vehicle-year) £103.9M; Royalties £64.1M; Engineering effort (module catalogue) £49.6M.

7. Supplier quote check

Line	Should-cost	Quote	Gap
No quote entered on the Supplier check tab			

Negotiation target: a fair Tier-1 price for this scope is £69.2M of engineering at £10,267 per person-month (should-cost re-costed as a Tier-1 supplier: its effort factor ÷ (1 - 12 % margin)).

8. Published programme figures (unverified)

Vehicle / Programme	Total SW Cost	£/Vehicle	Source
BMW iX (2021–2026)	£620.0M	£4,800	UNVERIFIED — Named as "Berylls Strategy Advisors estimate, 2023" — not found
Porsche Taycan (2019–2024)	£480.0M	£5,200	UNVERIFIED — Named as "SBD Automotive teardown + SW analysis" — no link
Mercedes EQS (2021–2026)	£710.0M	£5,500	UNVERIFIED — Named as "analyst estimate" / "McKinsey 2022" (two names in two files) — no link
Range Rover (L460, 2022–2027)	£390.0M	£3,800	UNVERIFIED — Named as "JLR programme estimate" / "JLR investor reports" (two names) — no link
Tesla Model S (Gen 3 HW4)	£850.0M	£3,200	UNVERIFIED — Named as "Morgan Stanley Research" — not found
Audi Q8 e-tron (2023–2028)	£520.0M	£4,600	UNVERIFIED — Named as "VW Group Annual Report + EY SW cost model" — no link
Lucid Air (2022–2027)	£380.0M	£7,800	UNVERIFIED — Named as "Lucid investor notes" — no link
This programme	£304.6M	£254	CostVision model — this calculation